HWA CHONG INSTITUTION
JC2 Preliminary Examination
Higher 2

CANDIDATE NAME		**CT GROUP**	
CENTRE NUMBER		**INDEX NUMBER**	

GEOGRAPHY

9173/02

Paper 2

17 September 2026

3 hours

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

Section A.
Answer **Question 1**.

Section B.
Answer **Question 2**.

Section C.
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

Submit your answers in <u>two separate sets</u>:
1. **Questions 1 and 3**
2. **Questions 2 and 4**

Answer Questions 1 and 2 and
EITHER Question 3 **OR** Question 4

Section A

Cluster 4 Fieldwork

1 A group of four Geography students in Singapore investigated the effectiveness of sheltered walkways as an adaptation strategy for residents coping with rising temperatures. They focused on a route between an HDB residential block next to their school and a nearby Mass Rapid Transit (MRT) station.

The students noted that while a sheltered walkway existed, it was incomplete as the walkway did not cover the entire route to the MRT station. Instead of walking to the MRT station from their homes, many residents continued to take a bus to the MRT station.

The students came up with the following geographical question for their fieldwork:

"Why do residents take a bus to the MRT train station instead of using the walkway?"

They divided themselves into two pairs, and conducted their fieldwork from 4 to 6 pm, every Wednesday in May as it was their designated home-based learning (HBL) day. Each day, one pair would count the number of residents using the sheltered walkway to get to the MRT station, while another pair would count the number of residents alighting from the bus at the MRT station. The students also noted the weather and traffic conditions.

In addition, they used a digital temperature sensor to measure the ambient temperature at seven selected points along the route. They also noted the features and details of the immediate surroundings at each point. They did this at 2 pm on a sunny Thursday afternoon in June.

Resource 1A is a map showing the walking route and the selected temperature measurement points. Resource 1B shows a photograph of a section along the route between Measuring Points 6 and 7. Resource 2 shows the number of residents using the route in the month of May, together with the prevailing weather and traffic conditions on each day. Resource 3A shows the information collected at each selected point along the route in June. Resource 3B shows a picture of the digital temperature sensor.

(a) Explain why the students' question is appropriate for their investigation. [3]

(b) With reference to Resources 1A and 1B, explain how students can minimise potential risks when conducting their fieldwork along the walkway between their school and the MRT train station. [6]

(c) With reference to Resources 2 and 3A, explain one strength and one weakness of the data representation methods. [2]

(d) Suggest reasons for the number of residents walking to the MRT train station shown in Resource 2. [4]

(e) With reference to Resources 1A, 1B, and 3A, explain why the students' selection of the 7 temperature measurement points is appropriate. [5]

[Turn over

(f) With reference to Resources 3A and 3B, explain how the students can ensure the reliability of the ambient temperature data they had collected. [6]

(g) The students wish to conduct a survey to find out why residents are not using the sheltered walkway more. Suggest two pieces of data that the students can collect to contribute to the analysis of data. [4]

(h) The students concluded that residents do not use the sheltered walkway mainly because it did not fully cover the route between their homes and the train station. With reference to all the resources, evaluate the validity of the students' conclusion. [10]

Section B

Cluster 3 Sustainable Future and Climate Change

2 Resource 4A shows different mobility patterns between women and men in Vienna. Resource 4B shows different activities of the unpaid reproductive work performed by women and men in Vienna. Resource 5 shows the level of satisfaction of women living in Vienna with their living situation in 2022. Resource 6 shows goals for mobility and transport for the Smart City Strategy in Vienna. Resource 7 shows Vienna's measures for a 15-minute city.

(a) Describe the distribution of mobility patterns and activities of unpaid reproductive work between women and men shown in Resources 4A and 4B. [4]

(b) With reference to Resource 5, cite data to compare the level of satisfaction of women living in Vienna with the aspects of their living situation. [4]

(c) With reference to Resource 6, explain the complementarity between sustainable urban development and liveability. [4]

(d) Explain how gender equality contributes to progress towards liveable cities. [6]

(e) Suggest how the concept of a 15-minute city and micromobility shown in Resource 7 may address the needs of women in Vienna shown in Resources 4A, 4B and 5. [6]

(f) Explain why the success in urban reimaging strategies may vary across places. [6]

[Turn over

Section C

Answer EITHER question 3 OR question 4

Cluster 3 Sustainable Future and Climate Change

3 'Psychological well-being is the most challenging dimension to achieve when creating liveable cities for the elderly.'

Evaluate this statement. **[20]**

4 'It is impossible for effective climate change response to be unilateral in nature.'

Evaluate this statement. **[20]**

Acknowledgements
Question 1 Resource 1A @ Adapted from EduGIS
Question 1 Resource 1B @ Google Maps via Google Street View
Question 1 Resource 2 @ Adapted from https://datamall.lta.gov.sg/content/datamall/en/static-data.html#Road%20Infrastructure
Question 1 Resource 3A @ Adapted from https://datamall.lta.gov.sg/content/datamall/en/static-data.html#Road%20Infrastructure
Question 1 Resource 3B @ https://www.pasco.com/products/sensors/wireless/ps-3201
Question 2 Resource 4A @ https://www.tandfonline.com/doi/full/10.1080/02665433.2022.2074526#d1e136
Question 2 Resource 4B @ https://www.tandfonline.com/doi/full/10.1080/02665433.2022.2074526#d1e136
Question 2 Resource 5 @ https://feministeerium.ee/en/vienna-urban-planning/
Question 2 Resource 6 @ https://smartcity.wien.gv.at/wp-content/uploads/sites/3/2022/05/scwr_klima_2022_web-EN.pdf
Question 2 Resource 7 @ https://www.voi.com/blog/reimaging-cities-the-case-of-nestroyplatz-vienna